SOFTWARE ENGINEERING LAB REPORT

Data Flow Diagram

LAB 5 | Case Study: Food Delivery Application

Project Overview

The Food Delivery Application allows customers to discover restaurants, browse menus, place and pay for orders, track delivery, and provide feedback. Restaurants manage menus and incoming orders, while delivery partners accept assignments and update delivery status. An administrator supervises users, restaurants, payments, and operational issues.

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1. Aim

To represent how data moves between users, processes, and data stores in the Food Delivery Application.

2. System Scope

• Customer: manages profile, browses menus, submits orders, and receives status updates.

• Restaurant: maintains menu availability and accepts or rejects orders.

• Delivery partner: receives delivery assignments and updates pickup/delivery status.

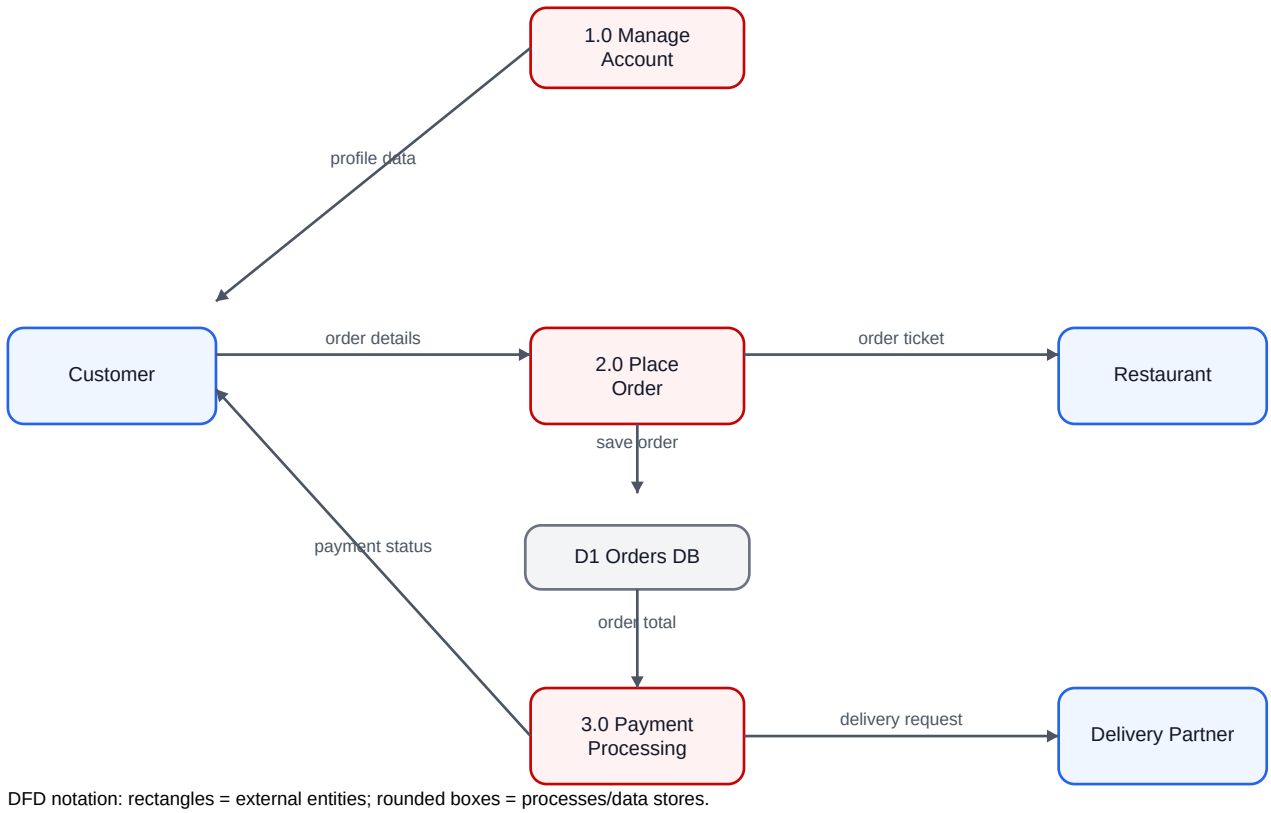
• Administrator: manages users, restaurants, disputes, and operational reports.

3. Context Diagram (Level 0)

The system exchanges customer/order information with customers, restaurant/order information with restaurants, delivery assignments with riders, and payment requests/status with a payment gateway.

External Entity	Data sent to system	Data received from system
Customer	Credentials, address, cart, order, feedback	Menu, price, order status, receipt
Restaurant	Menu updates, accept/reject, preparation status	New order details, customer delivery notes
Delivery Partner	Availability, pickup and delivery status	Assignment, address, navigation details
Payment Gateway	Payment confirmation/failure, transaction reference	Payment request, amount, payment token

4. Level 1 Data Flow Diagram



DFD notation: rectangles = external entities; rounded boxes = processes/data stores.

5. Main Data Stores

• D1 Users: customer, restaurant, rider, and administrator profiles.

• D2 Menus: restaurants, categories, items, prices, and availability.

• D3 Orders: order lines, totals, addresses, status history, and timestamps.

• D4 Payments: transaction reference, amount, status, and refund records.

• D5 Reviews: ratings, comments, and moderation status.

6. Result

The DFD identifies key external entities, processes, and data exchanges. Payment-card data should be handled by a compliant payment provider; the application should store only permitted tokens and transaction references.